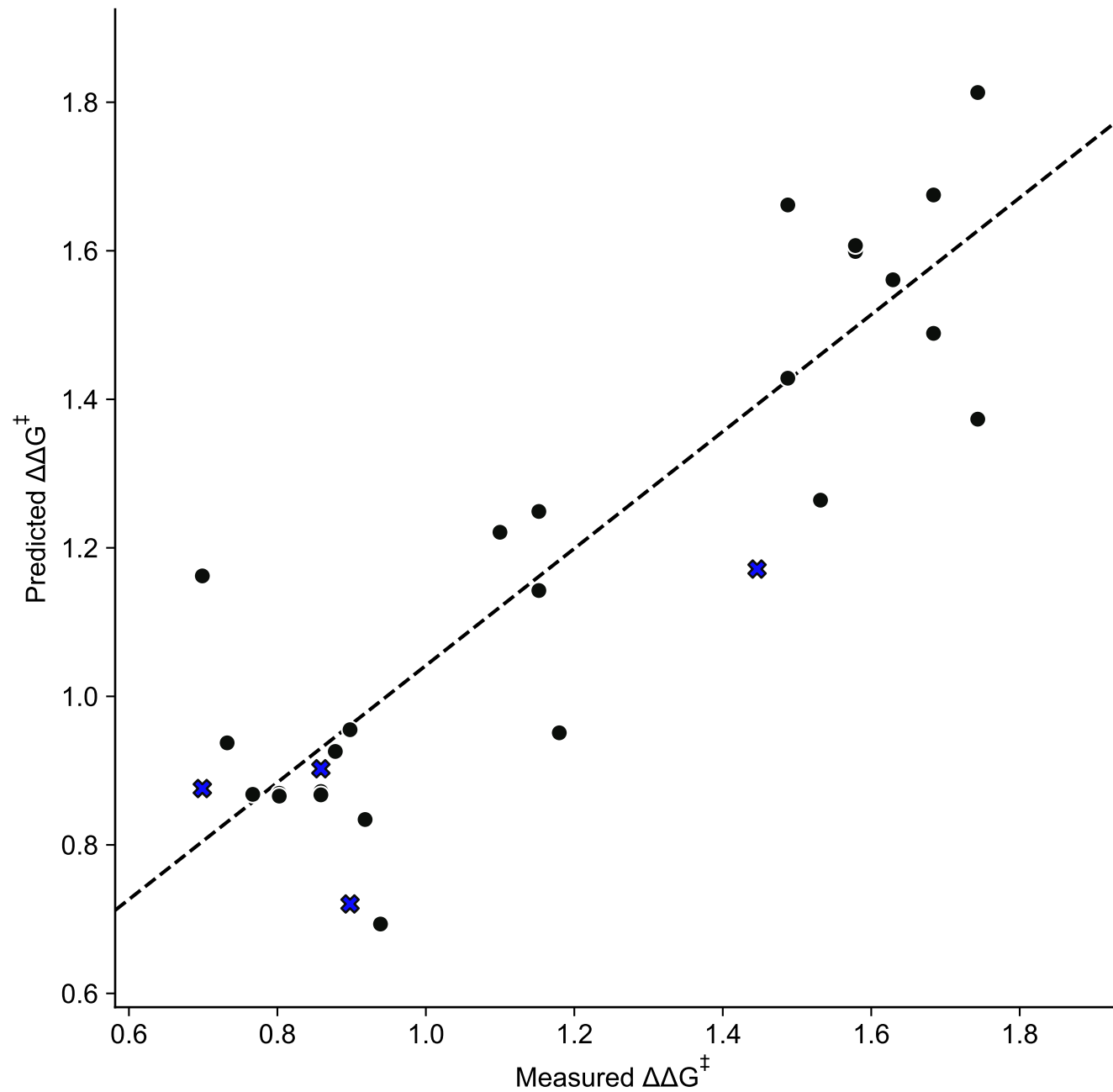




Model Metrics

Metric	Value
R ² (train)	0.7874
adj R ² (train)	0.7570
MAE (train)	0.1230
Q ² 3-fold	0.6530
MAE 3-fold	0.1646
Q ² 5-fold	0.7097
MAE 5-fold	0.1529
Q ² LOOCV	0.7292
MAE LOOCV	0.1412
R ² (ext. val.)	0.5587
adj R ² (ext. val.)	nan
MAE (ext. val.)	0.1683

Coefficient Estimates

Estimate	Std. Error	t value	p value
1.157	0.037	30.917	0.0
-0.254	0.055	-4.646	0.0
0.235	0.039	6.068	0.0
0.36	0.047	7.643	0.0

Cross-Validation Metrics

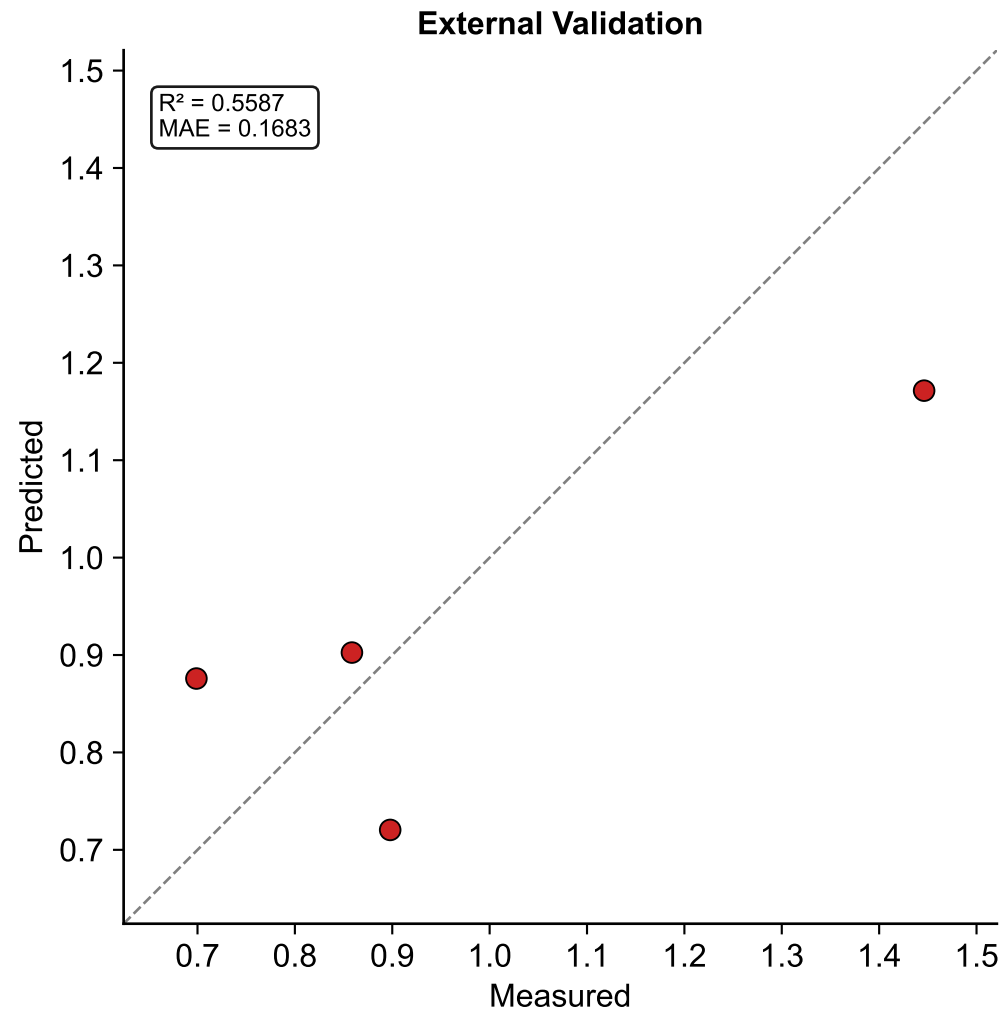
Metric	3-Fold	5-Fold	LOOCV
Q²	0.653	0.7097	0.7292
MAE	0.1646	0.1529	0.1412
RMSD	0.2144	0.1977	0.1911

Variance Inflation Factors (VIF)

variable	VIF
nbo_atom_5	1.761
dihedral_[5, 1, 8, 6]	1.547
B1_B5_angle_4-2	1.228

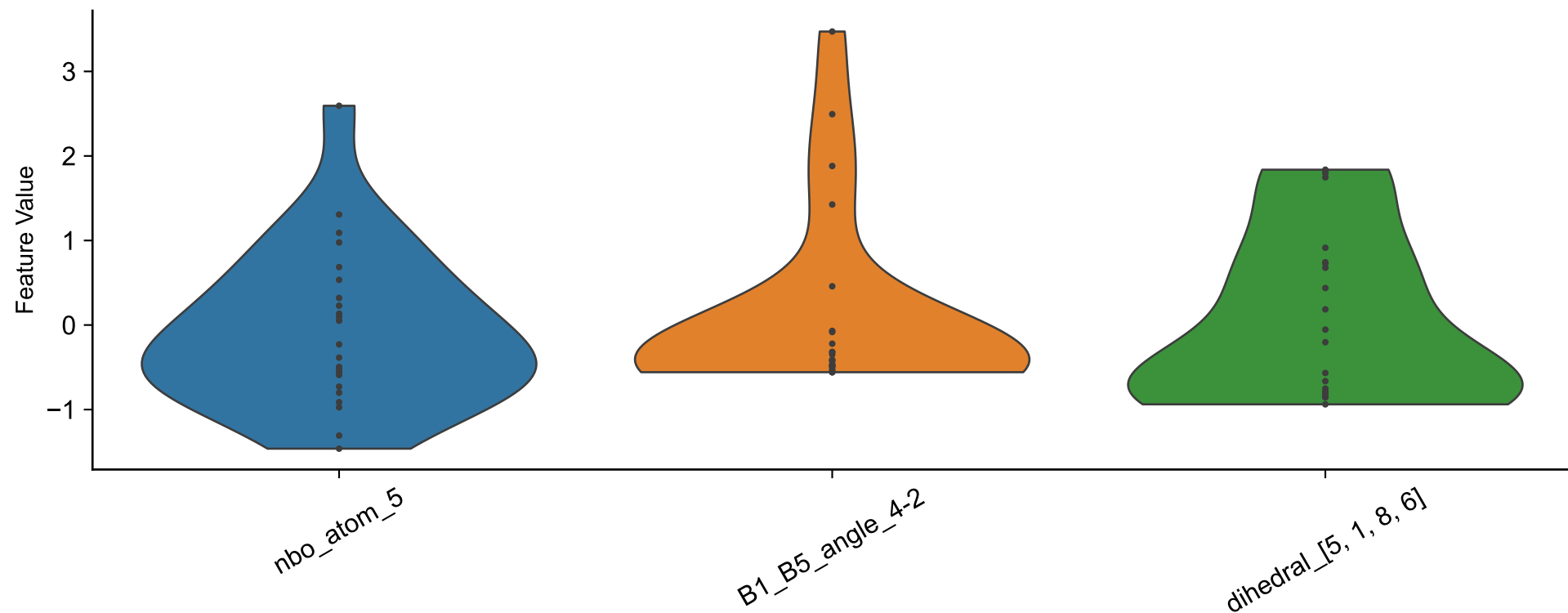
Coefficient Estimates

Estimate	Std. Error	t value	p value
1.1571	0.0374	30.9175	0.0
-0.2541	0.0547	-4.646	0.0001
0.2351	0.0388	6.0675	0.0
0.3597	0.0471	7.6431	0.0



Molecule	Actual	Predicted	Error in %
L10	0.898	0.72	19.761
L27	1.446	1.171	19.0
L21	0.699	0.876	25.324
L18	0.859	0.903	5.117

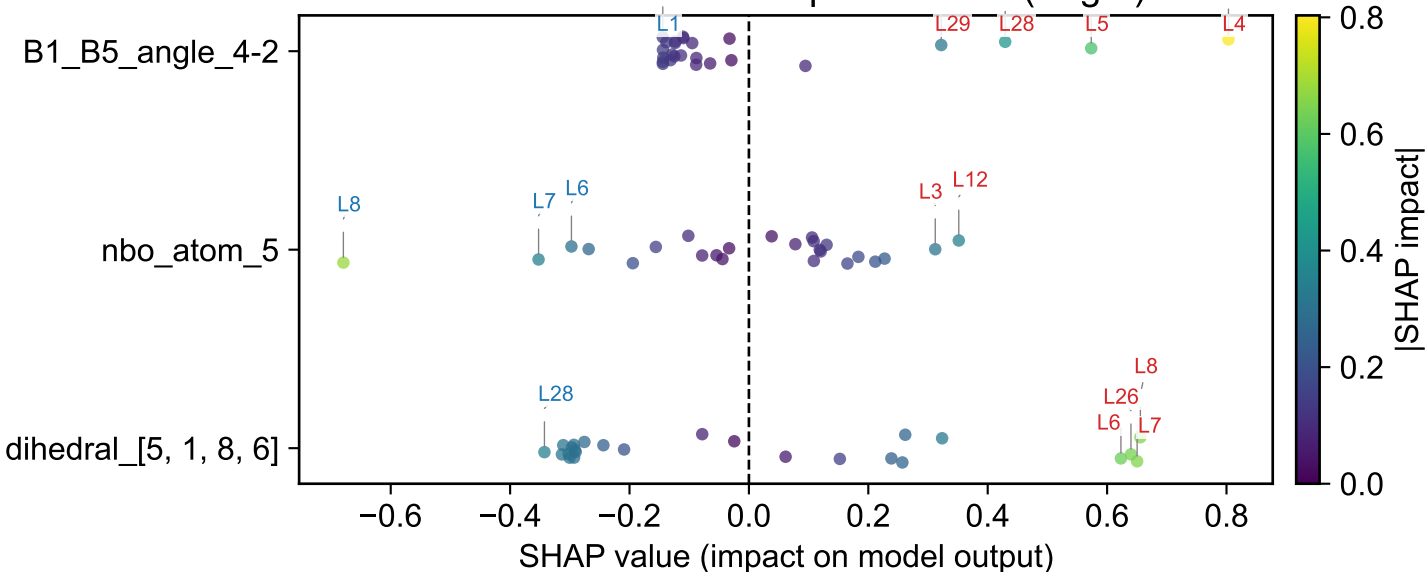
Distribution of Selected Features



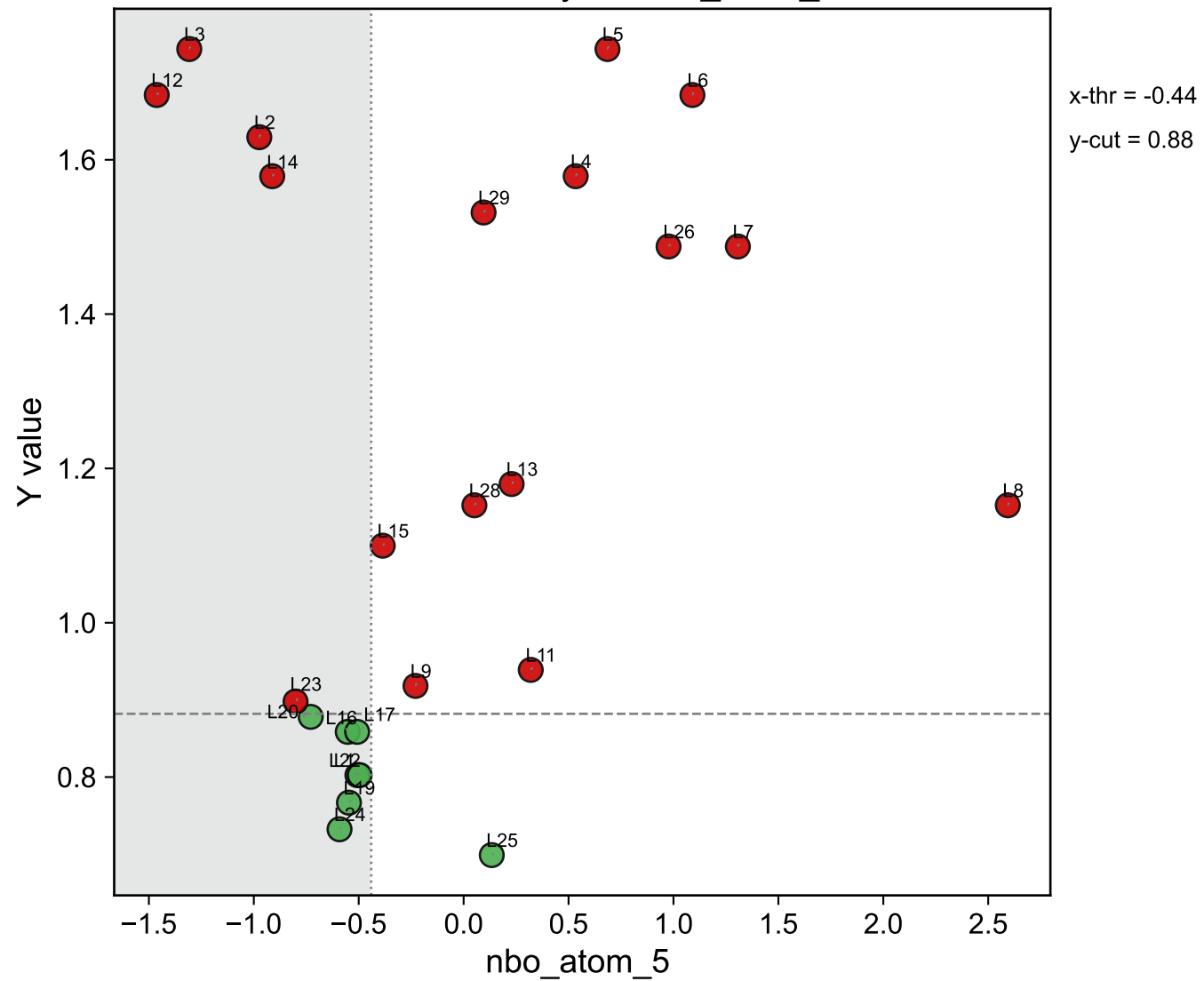
Descriptive Statistics

Feature	mean	std	min	25%	50%	75%	max
nbo_atom_5	-0.079	0.915	-1.463	-0.591	-0.386	0.32	2.593
B1_B5_angle_4-2	0.056	1.078	-0.558	-0.503	-0.414	-0.083	3.472
dihedral_[5, 1, 8, 6]	0.014	0.996	-0.938	-0.801	-0.567	0.729	1.837

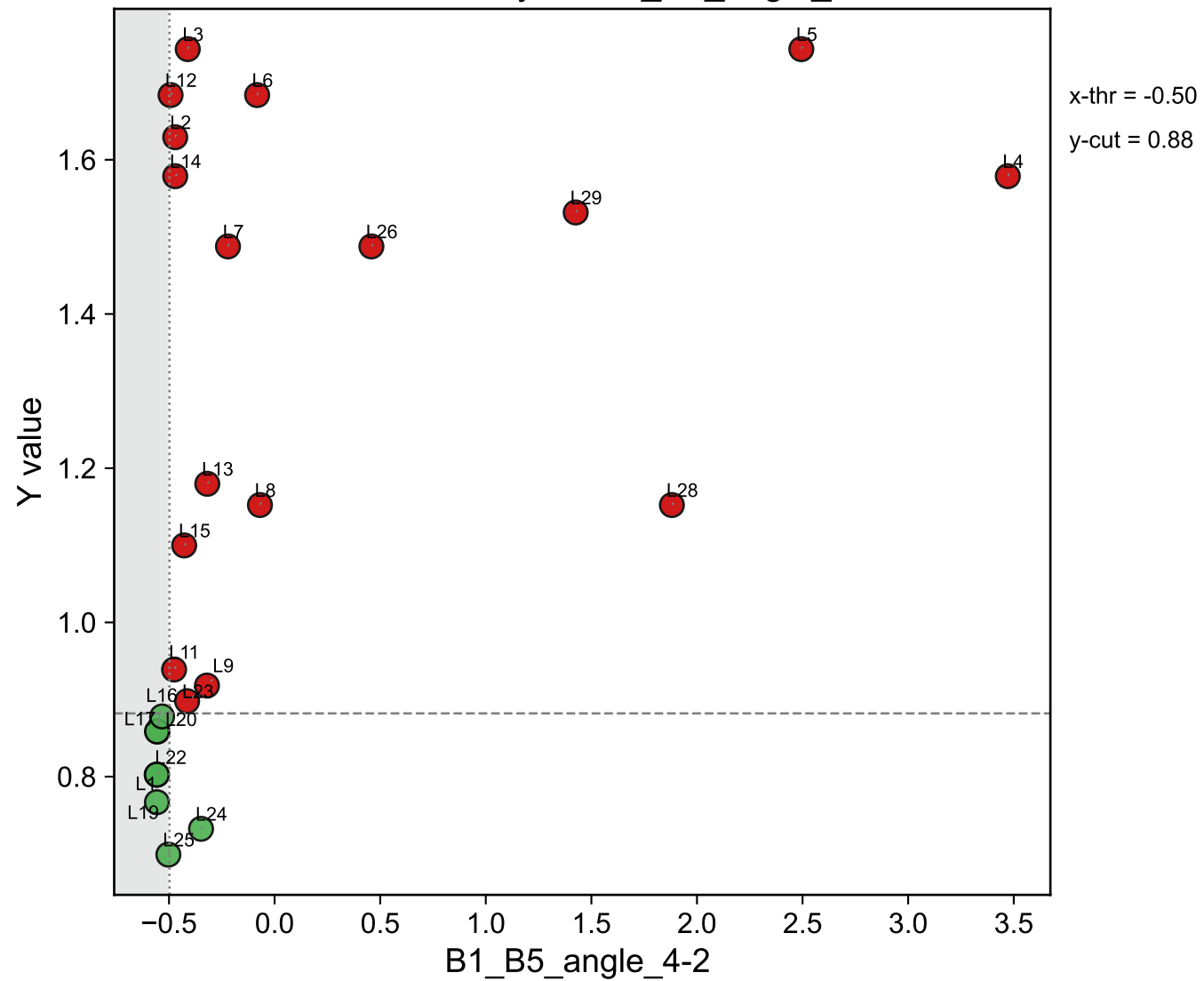
SHAP beeswarm — top 3 features (target)



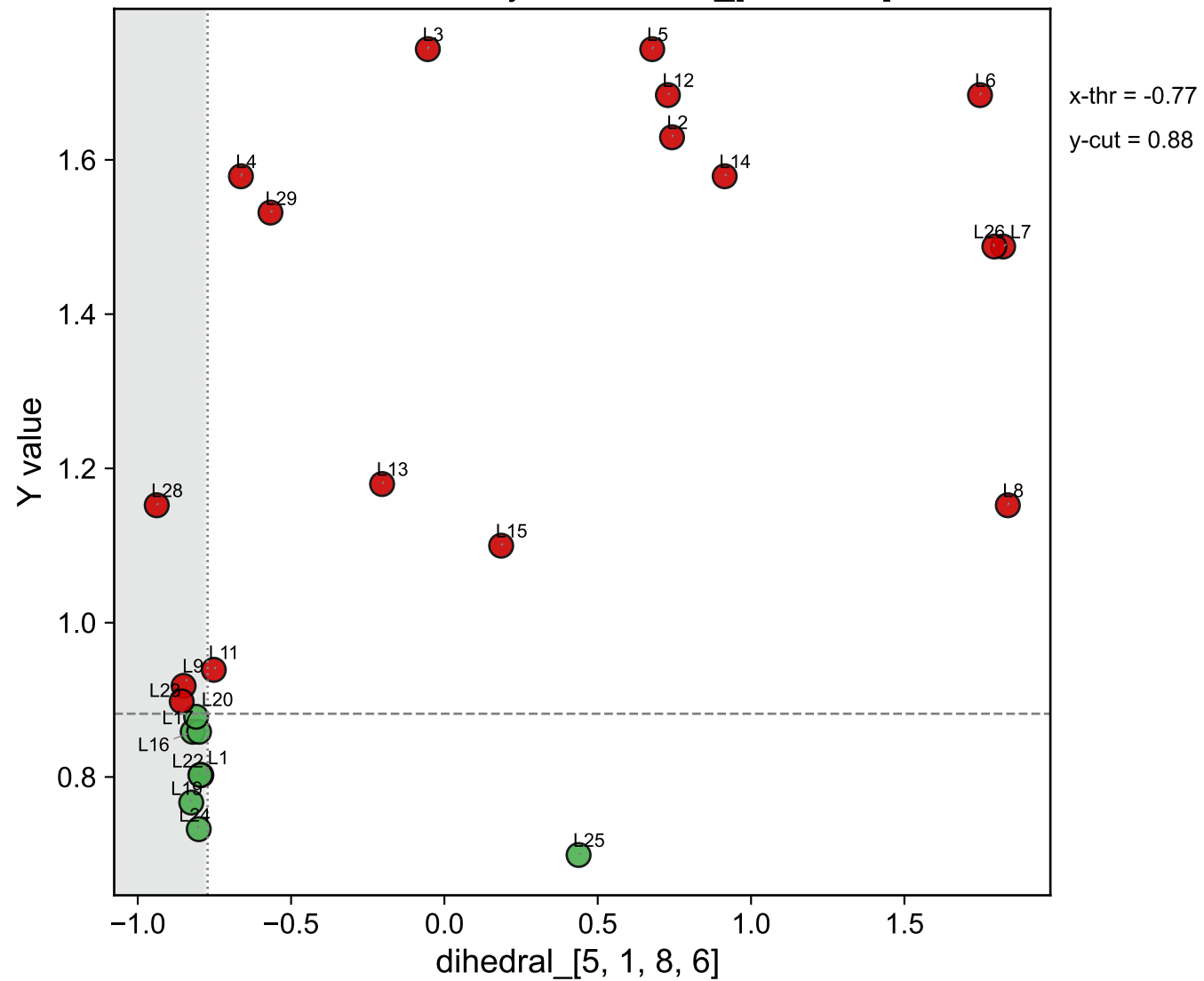
Threshold analysis: nbo_atom_5



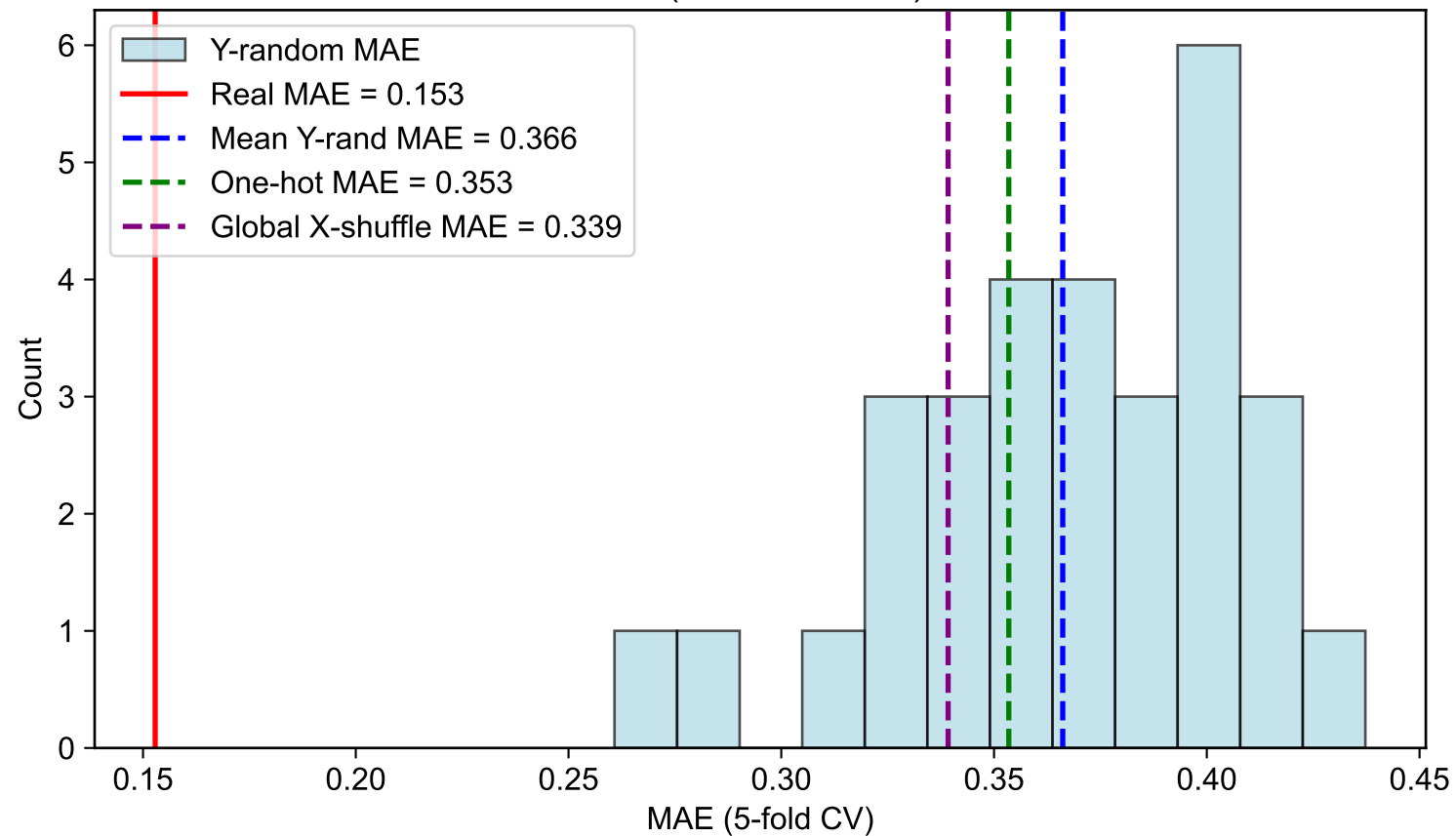
Threshold analysis: B1_B5_angle_4-2



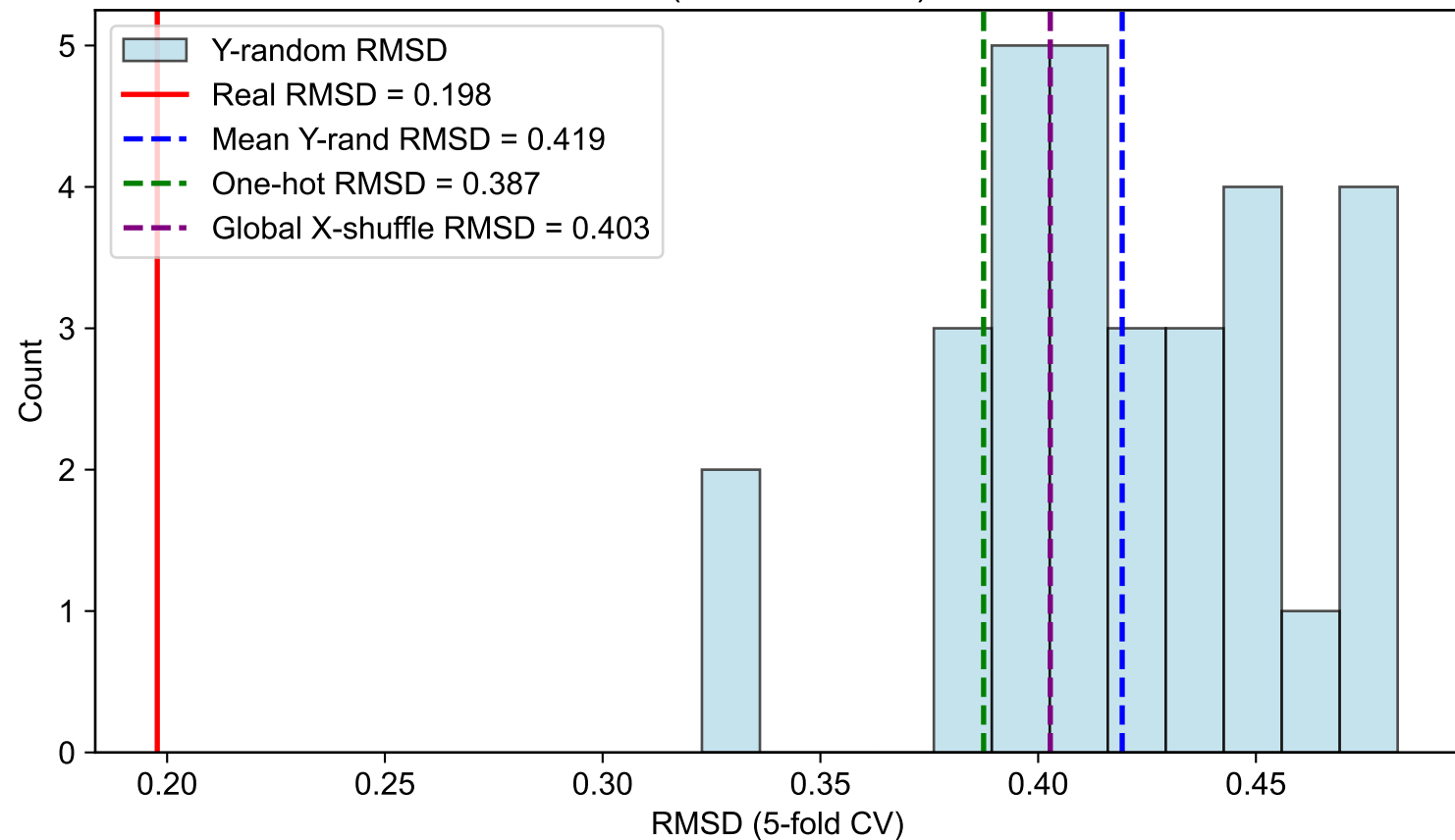
Threshold analysis: dihedral_[5, 1, 8, 6]



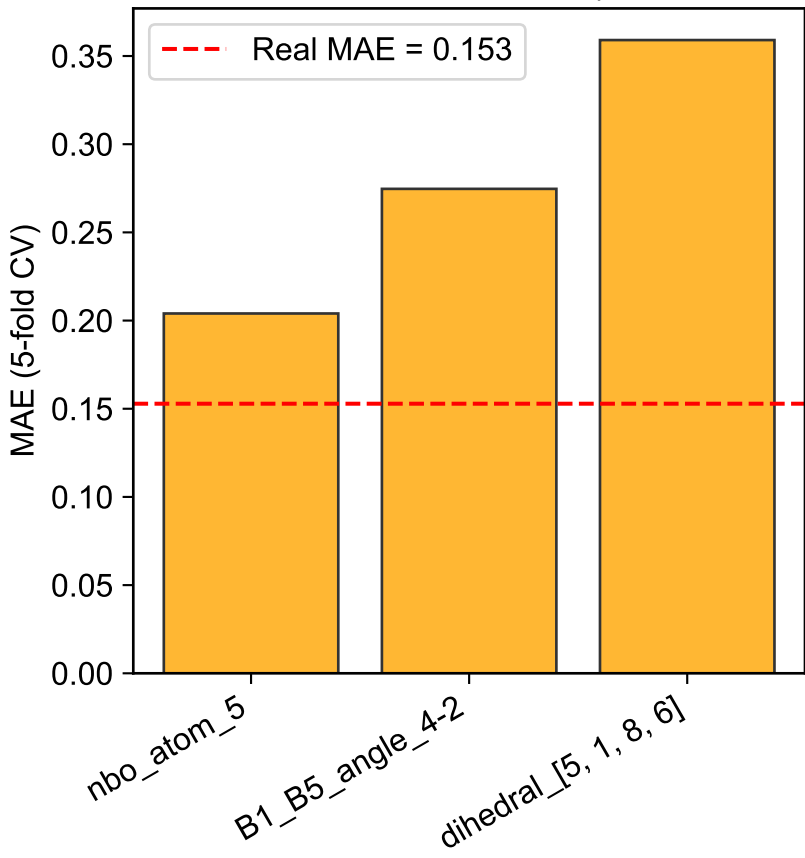
Validation (lower is better): MAE



Validation (lower is better): RMSD



Per-feature shuffle test: MAE (lower is better)



Per-feature shuffle test: RMSD (lower is better)

